

CHALLENGE BOOKLET

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The Intelligent Enterprise

Design and build a delivery planning system for Waypoint Group, then develop the predictions that help its team plan ahead. This brief sets out the business problem, phase requirements, datasets, judging criteria, and submission deadlines.

How the competition works

Tech-Triathlon 2026 follows one business challenge across three phases over 15 days. We release this brief and all datasets on Day 1. There are no separate challenge releases. All dates and times in this brief use Sri Lanka time (Asia/Colombo, UTC+05:30).

Milestone	Day	Date and time in Sri Lanka
Brief and all datasets released	1	Friday, September 25, 2026, at 12:01 AM
Designathon deadline	5	Tuesday, September 29, 2026, at 11:59 PM
Hackathon deadline	10	Sunday, October 4, 2026, at 11:59 PM
Datathon deadline	15	Friday, October 9, 2026, at 11:59 PM

Your Hackathon build must follow your Designathon submission. Judges will assess the continuity between the two. If you miss a phase, you may continue to the next, but you will receive zero for the missed phase. All three phases contribute equally to your overall score.

Waypoint Group

Waypoint Group (Pvt) Ltd is a fictional Sri Lankan retail group with three brands sharing one distribution network.

Brand	Outlets	Goods	Delivery schedule
Waypoint Fresh	80	Groceries, chilled and frozen goods; a wide range of products	Daily; before stores open at 8 AM
Waypoint Style	25	Hanging garments and cartons	Weekly, with seasonal peaks
Waypoint Tech	15	Appliances and consumer electronics	As needed; high-value, fragile goods

The network serves 120 outlets through a distribution center in Peliyagoda and a regional hub in Kandy. Its 60 vehicles include 12 refrigerated trucks, 40 dry-box trucks, and eight small vans for outlets that larger vehicles cannot reach. Four of the vans are refrigerated, giving the fleet 16 vehicles that can carry chilled goods. Each vehicle operates from its assigned depot.

The business problem

Waypoint's three brands compete for the same delivery capacity. Fresh needs deliveries to reach its 80 supermarkets before they open at 8 AM. Chilled orders require refrigerated vehicles, and some outlets can only be reached by van. Style's garments fill a vehicle's available volume before reaching its weight limit, and around half of its stores are in malls with fixed delivery windows. Tech's appliances are heavy, fragile, and valuable, with demand that varies from day to day.

On most days, the fleet cannot meet every brand's needs at once. Dispatchers must allocate capacity while accounting for outlet access, delivery windows, and weekly fuel quotas. When demand exceeds capacity, they must decide which orders to defer and explain the consequences.

How orders reach the dispatcher

Store managers place orders according to their brand's delivery schedule. Fresh outlets order dry groceries for every operating day they trade and place separate chilled orders on several days each week. A Fresh outlet can therefore have two orders for the same delivery day. Style orders weekly for a scheduled delivery day, with larger orders ahead of seasonal peaks. Tech orders as needed, often for a single large item.

Orders for the next day close at 4 PM. After the cutoff, the dispatcher plans against the confirmed orders, available vehicles, and operating constraints. Orders received after the cutoff wait for the following run.

How Waypoint works today

Dispatchers plan deliveries using spreadsheets and their network knowledge. They communicate instructions through phone calls, conversations at the loading dock, and printed run sheets. Once vehicles leave, dispatchers have no shared view of progress. Drivers report problems by phone, and changes reach each person through separate calls. Handwritten notes provide only a limited record of deliveries and deferral decisions.

Problems your solution must address

- **Planning is fragmented.** Orders arrive by phone or message and are entered again in a spreadsheet. The plan depends on one dispatcher's knowledge.
- **Delivery progress is difficult to track.** Dispatchers usually learn about a problem only after a driver has reached the outlet.
- **Deferrals lack a clear record.** Decisions made under pressure can leave the same outlet unserved on consecutive runs.
- **Communication does not support feedback.** Printed run sheets and verbal instructions provide no reliable way to record proof of delivery or flag a loading shortfall before departure.
- **Demand is difficult to anticipate.** Waypoint cannot estimate the vehicles, drivers, or refrigerated capacity it will need ahead of payday and festivals.
- **Service time and lateness are not predicted.** Dispatchers discover delays after they have affected a delivery.
- **Field connectivity is unreliable.** The solution must support work without a connection and reconcile records when connectivity returns.

Operating constraints

Vehicles

- Every vehicle has a weight limit and a volume limit. A load must satisfy both.
- Only refrigerated vehicles may carry chilled or frozen goods. Refrigerated vehicles may also carry ambient goods. Ambient vehicles cannot carry chilled or frozen goods.
- Each vehicle has a weekly fuel quota. Route distance consumes that allowance.
- A vehicle can run up to two routes per day. Waypoint operates Monday through Saturday.
- Each vehicle has a driver. Driver availability is not a separate constraint when allocating the existing fleet.

Outlets

- Every outlet has a delivery window. Fresh deliveries must arrive before stores open at 8 AM, although individual outlets' windows may differ.
- Mall outlets accept deliveries only within the mall's fixed access window.
- Outlets marked `van_only` cannot be served by trucks.
- Unloading conditions vary by outlet. Goods may arrive through a rear dock, at the curb, or through a shared mall loading bay.

Demand and operating days

Paydays, festivals, weekends, and monsoon conditions affect demand or travel time. Use `calendar.csv` to identify operating dates.

When demand exceeds capacity, the dispatcher must decide which orders move to the next run and record the reason.

Connectivity

Mobile coverage can drop across hill country, the Kandy corridor, and rural districts.

Work away from the depot must remain usable offline. Records must reconcile when the connection returns.

The four user roles

Design the system around the conditions each person works in and the information they need from other roles.

Dispatcher

Works at a large screen in the Peliyagoda planning office with stable connectivity.
Builds the daily plan using a spreadsheet and knowledge of outlet restrictions and vehicle capabilities.

Needs visibility into delivery progress and problems after vehicles leave the depot.
Needs to explain deferral decisions and identify outlets that have already been skipped.

Loader

Works at the Peliyagoda or Kandy warehouse dock using a shared tablet or terminal. Printed loading lists can become outdated when plans change.

Needs the stop sequence so goods can be loaded in an order that supports unloading.
Needs to flag missing or damaged items before a vehicle leaves.

Driver

Works on the road using a personal phone. Currently relies on a paper run sheet and phone calls for changes. Design interactions for use when safely stopped.

Needs to record delivery outcomes and proof of delivery so disputes do not depend on memory.
Needs to record work offline when coverage drops and synchronize it when connectivity returns.

Store manager

Works at the outlet counter using a desktop or phone. Places orders by phone or message without confirmation that they received or scheduled them.

Needs an expected arrival time to schedule staff to receive goods.
Needs clear notice when an order is deferred, plus a way to confirm receipt and report issues.

Your objective

Build a system that connects ordering, planning, loading, delivery, and receipt across these four roles. Help Waypoint make delivery decisions it can explain and plan capacity ahead of demand. The Designathon defines the experience, and the Hackathon implements it. The Datathon develops estimates of outlet service time, arrival lateness, and future demand volume to support planning.

The workflow below shows what each role needs to do. Service-time and lateness predictions support delivery planning. Demand forecasts support fleet planning for future weeks.

Stage	Role	System requirement
Place order	Store manager	Capture and confirm the order before the cutoff.
Close orders	Dispatcher	Bring confirmed orders into one queue.
Plan and allocate	Dispatcher	Assign served orders to vehicles and trips; identify deferred orders.
Load	Loader	Load for the planned stop sequence and flag shortfalls.
Deliver	Driver	Follow the route and record each stop, including while offline.
Confirm receipt	Store manager	Confirm what arrived and report issues.
Plan future capacity	Dispatcher	Use demand forecasts to plan vehicles, drivers, and refrigerated capacity.

Shared datasets

All three phases use the same 120 outlets, 60 vehicles, two depots, and calendar. Use these records consistently across your designs, working system, and models. All competition data is synthetic and does not represent a real company, outlet, or person.

File	Purpose
<code>outlets.csv</code>	The 120 outlets, including brand, district, depot, access restrictions, and delivery windows.
<code>vehicles.csv</code>	The 60 vehicles, including type, temperature capability, weight and volume limits, fuel profile, and home depot.
<code>calendar.csv</code>	Dates across the history and forecast horizon, with payday, festival, monsoon, and operating-day information.

Additional Datathon files and their key columns are documented later in this brief.

[Click Here to Access the Datasets](#)

DESIGNATHON CHALLENGE

Designathon

Submission due Day 5

Design one system that helps all four roles complete the delivery workflow. Show how you understand the operation, which problems you prioritize, and why your scope is appropriate.

Connect the role-specific experiences. A dispatcher's decision should reach the loader, and a driver's delivery record should give the store manager information they can act on.

Scope

Design the screens each role needs to complete its work, including at least one fully developed screen for a failure scenario. Explain your choices. Judges will assess prioritization and restraint, as well as the quality of the experience.

Failure scenarios

Design at least one screen for a situation in which the normal workflow breaks down. This is called a degradation screen in the judging criteria. Choose the scenario, name it, and explain in a short paragraph why it matters to Waypoint. You may include more than one, but response quality matters more than the number of scenarios.

Deliverables

- **User personas.** Include one persona for each role, grounded in the working conditions and needs in this brief.
- **Screen flows.** Show each role's screens and include a one-paragraph rationale for every screen, explaining its purpose and priorities.
- **Degradation screens.** Include at least one fully designed failure scenario, with its name and rationale.
- **High-fidelity prototype.** Use a design tool of your choice to demonstrate the flows you designed.
- **Demo Video.** Upload a three to five minute demo video on YouTube as an unlisted video. The video should provide a walkthrough of your design workflow and discuss any assumptions that influenced your design for the given scenario. Submit the YouTube URL to the submission form.
- **AI tool disclosure.** Explain which work was AI-assisted, which was not, and how you used the tools.
- **Core tradeoff explanation, optional.** Use up to one page or one diagram to explain your main design tradeoff.
- **Style guide, optional.**

Judging criteria

Criterion	Weight
Problem framing	25%
Understanding of user context	20%
Degradation screen quality	15%
Domain accuracy	10%
Scope and prioritization	15%
Visual and interaction design, including consistency across roles	15%

Submission

Organize your personas, screen flows, rationale, degradation screens, diagrams, AI tool disclosure, core tradeoff explanation (optional), and style guide (optional) into one design file with distinct pages. Export the file using `TeamName_Designathon` as the base filename. Compress it as `TeamName_Designathon.zip` and upload it through the submission form. Upload shareable links to your prototype and demo video onto the submission form as well.

Submit by **Tuesday, September 29, 2026, at 11:59 PM** Sri Lanka time (Day 5). Judges will assess the design submitted at that deadline. You may refine the solution during the Hackathon, but document significant departures from the submitted design in your README.

Deadline for submissions: **Tuesday, September 29, 2026, at 11:59 PM**

🔗 **Submission Form:** <https://forms.gle/H6dqUZP6pXdGC8Go8>

HACKATHON CHALLENGE

Hackathon

Submission due Day 10

Build the flows, screens, and capabilities in your Designathon submission. That design is your implementation specification, and judges will assess how faithfully you deliver it.

Every submission must meet the following requirements.

Provide a responsive web application that lets a judge complete the delivery workflow across all four roles, from planning through loading and delivery to receipt at the outlet. Judges will assess the driver and loader experiences on phone-sized screens. Native applications are optional additions to the required web application.

Make the system respect the operating constraints. Plans must account for capacity, temperature requirements, outlet access, delivery windows, and fuel quotas.

Planning and allocation

Your system must assign orders to vehicles and trips and handle a day when demand exceeds available capacity. You may use automatic allocation, assisted planning, or manual decisions with validation. Whatever approach you choose, the system must produce an allocation that respects the operating constraints and identifies deferred orders.

Judge walkthrough

Add a numbered walkthrough to your README that a judge can follow across all four roles, from planning to completed delivery. Seed the system with the shared datasets and at least one realistic delivery day so the walkthrough works on a fresh installation.

Deliverables

- **Deployed system.** Provide a public URL and credentials for four seeded accounts, one per user role.
- **Source repository.** Use a GitHub monorepo named `TeamName_SolutionName`. It must contain:
 - A README with setup and configuration instructions, seeded account details, the judge walkthrough, and significant departures from the Designathon submission.
 - A Docker Compose file and an `.env.example` file at the repository root. The command `docker compose up` must start the complete stack, including the database and seed data.
 - A `docs` folder at the repository root containing an architecture diagram and data model, showing the main components and how the system stores and connects its data.
 - An AI tool disclosure in the same docs folder, explaining which work was AI-assisted, which was not, and how you used the tools.
- **Demo video.** Submit an unlisted YouTube video lasting five to eight minutes. Show all four roles completing the walkthrough, followed by a brief explanation of the code and architecture.

Judging criteria

Criterion	Weight
Functional completeness across all four roles	20%
Planning and allocation engine	20%
Degradation, offline operation, and recovery	10%
Fidelity to the Day 5 design	10%
Engineering quality and architecture	25%
Creativity	5%
Demo video	10%

Submission

Submit the repository link, deployed URL, seeded account credentials, and demo video link through the submission form.

Submit by **Sunday, October 4, 2026, at 11:59 PM** Sri Lanka time (Day 10). Code pushed after the deadline will not be considered. Keep the deployment live throughout the review period and, if your team advances, through the semifinal and Grand Finale periods.

Deadline for submissions: **Sunday, October 4, 2026, at 11:59 PM**

🔗 **Submission Form:** <https://forms.gle/WurHAKjbq2XEZQhbA>

DATATHON CHALLENGE

Datathon

Submission due Day 15

Waypoint needs estimates it can use before a delivery starts or demand peaks. Route records can show when a vehicle departed, how long it traveled, when it arrived, and when unloading finished. Order history can show how demand changes over time. Use the supplied synthetic records to develop predictions that support delivery planning.

The Datathon challenge is judged separately from the Hackathon system. Teams are not required to integrate their Datathon solutions into their Hackathon build.

Complete the two prediction tasks and the peak-day allocation task below.

Task 1: Predict service time and lateness

Your first challenge is to predict two things for each planned order (`delivery_id`) in the test input:

- `pred_service_min` - this denotes predicted handling time at the outlet, in minutes. Service time is how long the delivery will take to handle at the outlet (its service time, in minutes)
- `pred_late_prob` - the probability it runs late, meaning it arrives after the outlet's delivery window has closed.

Neither target is provided as a label in the input table. Construct the training labels from the dataset by working out the relationships from the column descriptions and document your reasoning. Correct label construction is part of the assessment.

Account for the following when defining your labels:

- Outlets receive goods only within their delivery window. A vehicle that arrives early waits until the window opens.
- A late arrival is still delivered in the supplied scenario. However, receiving staff may have moved to other duties, and a Fresh outlet may miss morning sales. Lateness refers to arrival after the window closes.
- At prediction time, planned departure, travel duration, and arrival are available for each delivery. Actual journey and handling times are available only in the training route records.

Inputs

Test `Data/task1_test_inputs.csv` - each row denotes a delivery plan for an order (`delivery_id`). Every `delivery_id` matches exactly one route leg.

Test `Data/route_legs_test.csv` - each row denotes a single leg of the route and contains the matching route legs, including origin, distance, and planned departure and arrival times.

Use relevant data from `Training Data/` and `General Data/` . Choose and justify your features. The challenge does not prescribe a feature set.

Outputs

Complete `Submission Templates/submission_task1.csv` . Keep the `delivery_id` values and row order exactly as supplied. Do not add or remove rows. Fill only the two prediction columns.

Column	Type	Required value
<code>delivery_id</code>	string	Supplied identifier. Keep unchanged.
<code>pred_service_min</code>	number	Predicted outlet handling time, in minutes.
<code>pred_late_prob</code>	number from 0 to 1	Probability of arrival after the outlet's delivery window closes.

Example output

<code>delivery_id</code>	<code>pred_service_min</code>	<code>pred_late_prob</code>
ORD0092308	16.4	0.07
ORD0092309	21.8	0.63

Task 2A: Forecast depot demand

Waypoint needs to know the demand in advance to plan vehicle, driver, and refrigerated capacity. Forecast the volume ordered for each depot and brand over the 10 future weeks in the test inputs.

Predict two values for every depot, brand, and week combination.

- `pred_total_volume_m3` is the total order volume for that combination, in cubic meters.
- `pred_chilled_volume_m3` is the chilled portion of that total, in cubic meters.

You only need to predict these volumes. You do not need to convert them into vehicle or driver requirements.

Build your training data from `deliveries_train.csv` and `task1_test_inputs.csv`. In both files, each row is one order, identified by `delivery_id`. Follow these rules when you prepare the data.

- Count every order once, including orders that were deferred or never dispatched. They still represent demand.
- Assign each order to the week the store requested the order.
- Use `iso_year` and `iso_week` from `calendar.csv` so the forecast periods match the supplied calendar.
- Only Fresh has chilled demand. Set `pred_chilled_volume_m3` to 0 for Style and Tech.

Inputs

Test Data/`task2a_test_inputs.csv` contains one row per depot, brand, and forecast week.

Use the `deliveries_train.csv` in Training Data, `task1_test_inputs.csv` in Test Data, and `calendar.csv` in General Data to construct your training data and features.

Outputs

Complete Submission Templates/`submission_task2a.csv`. Preserve the supplied `row_id` values and fill the two prediction columns.

Column	Type	Required value
<code>row_id</code>	string	Supplied identifier. Keep unchanged.
<code>pred_total_volume_m3</code>	number	Total demand volume for the depot, brand, and week, in cubic meters.
<code>pred_chilled_volume_m3</code>	number	Chilled volume within that total. Use 0 for Style and Tech.

Example output

<code>row_id</code>	<code>pred_total_volume_m3</code>	<code>pred_chilled_volume_m3</code>
W0000	412.7	151.3
W0001	96.4	0

Task 2B: Allocate the fleet on a peak day

Plan deliveries for one day when demand exceeds available fleet capacity. Produce the following outputs.

- A complete allocation that marks every order as “served” or “deferred” and assigns every served order to a vehicle and trip.
- A short written policy explaining your allocation priority and deferral decisions.

This task does not require a trained model. Analyze demand and available capacity, identify the limiting resources, and explain how your allocation responds. There is no single correct allocation. Judges will assess feasibility and the reasoning behind your decisions.

Scenario

Scenario S1 covers a single dispatch day at Peliyagoda. Orders have closed, and the available fleet is listed in the scenario files. You cannot use vehicles with status `in_workshop`. Allocate only vehicles marked as available.

Scenario	Depot	Conditions
S1	Peliyagoda	A festival is one week away. Fresh demand is rising, including dairy, meat, and produce. It is not a payday, and there are no monsoon conditions. Several vehicles are in the workshop.

Inputs

Test `Data/task2b_peak_day_scenarios.csv` contains every order, including its outlet, access restrictions, delivery window, temperature requirement, and size.

Test `Data/task2b_peak_day_fleet.csv` identifies the vehicles available on the scenario day and those in the workshop.

Use `General Data/vehicles.csv` for vehicle capacity, temperature capability, and home depot.

Use `General Data/district_travel.csv` and `General Data/service_allowance.csv` for the trip-time calculation.

Outputs

Complete `Submission Templates/submission_task2b.csv`. Keep `scenario`, `order_ref`, and `outlet_id` unchanged. Complete `decision`, `vehicle_id`, and `trip_id` for every row as specified below.

Column	Type	Required value
<code>scenario</code>	string	Supplied identifier; always S1.
<code>order_ref</code>	string	Supplied order identifier. Use this as the allocation key because <code>outlet_id</code> may appear more than once.
<code>outlet_id</code>	string	Supplied outlet identifier, included for readability.
<code>decision</code>	string	Served or deferred for every order.
<code>vehicle_id</code>	string	Assigned vehicle for a served order. Leave blank if deferred.
<code>trip_id</code>	1 or 2	Assigned trip for a served order. Leave blank if deferred. A vehicle can return to the depot and reload once, for a maximum of two trips. Orders sharing <code>vehicle_id</code> and <code>trip_id</code> form one trip and share its capacity and time limits. For example, VEH014 may carry four Colombo Fresh orders on trip 1 and three Gampaha Fresh orders on trip 2.

The three answer columns contain placeholders in the supplied template. Replace every placeholder. For deferred orders, leave `vehicle_id` and `trip_id` blank.

Template example

scenario	order_ref	outlet_id	decision	vehicle_id	trip_id
S1	S1-000	OUT001	(served/deferred)	(e.g. VEH014)	(1 or 2)
S1	S1-001	OUT001	(served/deferred)	(e.g. VEH014)	(1 or 2)
S1	S1-002	OUT002	(served/deferred)	(e.g. VEH014)	(1 or 2)

Completed example (illustrative only)

scenario	order_ref	outlet_id	decision	vehicle_id	trip_id
S1	S1-000	OUT001	served	VEH014	1
S1	S1-001	OUT001	served	VEH002	2
S1	S1-002	OUT002	deferred		

Feasibility rules

Your allocation must meet all rules below:

- 1 **Brand and district.** All orders sharing a `vehicle_id` and `trip_id` must belong to the same brand and district.
- 2 **Refrigeration.** Orders with `temp_requirement = chilled` require a vehicle with `temp = reefer`. Refrigerated vehicles may also carry ambient orders.
- 3 **Vehicle access.** Outlets with `parking_constraint = van_only` require a vehicle with `type = van`.
- 4 **Home depot.** A vehicle may serve only outlets assigned to its own depot.
- 5 **Whole orders.** Assign each served order to one vehicle and one trip. Do not split an order across trips or vehicles.
- 6 **Capacity.** For each trip, total `order_volume_m3` must not exceed `volume_cap_m3`, and total `order_weight_kg` must not exceed `weight_cap_kg`.
- 7 **Trips and time.** Each vehicle may run at most two trips in total. Its trips must fit the daily time budgets below.

Calculate trip time

Each trip leaves the depot, travels to one district, and delivers its orders there. For Task 2B, calculate the outbound journey, travel between stops, and handling time. Do not add the return journey to the depot; the stated budgets already allow for it. Calculate the trip time in three steps:

- **Step 1 Time for outbound travel:** Take `depot_to_district_freeflow_min` from the trip's district row in `district_travel.csv`. Count it once per trip.
- **Step 2 Time for travel between stops:** Multiply `inter_stop_freeflow_min` for that district by the number of orders minus one. A trip with three orders has two inter-stop journeys; a trip with one order has none.
- **Step 3 Handling time:** For each order, look up `service_allowance_min` in `service_allowance.csv` using the trip's brand and the outlet's `dock_type`. Add the allowances for all stops.

Add the three components to calculate trip duration.

$$\text{trip_minutes} = \text{outbound travel} + \text{inter-stop travel} + \text{total handling time}$$

Example: A Fresh trip to Gampaha carries three orders, two with rear docks and one with street access.

Component	Reference and calculation	Minutes
Depot to Gampaha	<code>depot_to_district_freeflow_min</code> = 37	37
Travel between three stops	<code>inter_stop_freeflow_min</code> = 9; $9 \times (3 - 1)$	18
Handle stop 1	Fresh + rear_dock	15
Handle stop 2	Fresh + rear_dock	15
Handle stop 3	Fresh + street	16
Trip total	37 + 18 + 15 + 15 + 16	101

Apply the following budgets to each vehicle's daily trips.

Trips	Operating window	Daily budget per vehicle
Fresh	3:30 AM to 8 AM	270 minutes
Style and Tech combined	Trading day	480 minutes

The Fresh budget applies to the total time for that vehicle's Fresh trips. The Style and Tech budget applies to the combined time for its Style and Tech trips. These are separate windows. A vehicle may run one Fresh trip and one Style trip, checked against their respective budgets, but may still run only two trips in total.

For example, a second Fresh trip to Colombo with four street-access stops takes $24 + (3 \times 8) + (4 \times 16) = 112$ minutes. Combined with the Gampaha trip, the vehicle uses $101 + 112 = 213$ of its 270 Fresh minutes. A third trip is not allowed.

Written prioritization policy

Submit a write-up of approximately one page or less. Show the calculations behind your allocation and explain why you deferred specific orders. Your analysis should show the calculations behind your allocation, identify what limited service on this day, and explain which deferrals were unavoidable and which were your choice, and what they cost.

Rules and Regulations

- **Deadline:** The submission form closes after the deadline.
- **Model Restrictions:** You are restricted from using any pre-trained models, except for synthetic data generation or pre-processing.
- **API Usage:** Proprietary API-based modelling/preprocessing is prohibited.
- **Integrity:** Cheating, plagiarism, or rule violations will result in disqualification.
- Usage of Low-code/No-code AI tools or fully automated end-to-end modelling tools are strictly prohibited.

Terms and Conditions

- **Use of Data:** The provided datasets may be used solely for the purpose of this competition. Any other use, including but not limited to commercial purposes, academic research, or personal projects, is strictly prohibited.
- **Data Sharing:** The datasets must not be shared, distributed, or transmitted in any form – whether publicly or privately – to any third party. This includes uploading the datasets to external websites, forums, or social media platforms.
- **Publication and Disclosure:** You are not permitted to publish, disclose, or make the datasets or any derivatives publicly available unless explicitly authorized by the competition organizers.
- **Data Confidentiality:** By participating in the competition, you agree to maintain the confidentiality of the datasets and any sensitive information contained within them.
- **Violation of Terms:** Any violation of these terms and conditions may result in disqualification from the competition.

Deliverables

- **Architecture diagrams.** Show your models, preprocessing pipeline, and proposed deployment approach. High-level diagrams are sufficient.
- **Data preprocessing document.** A brief write-up of your data preparation, label construction, data cleaning, feature engineering, and rationale.
- **Model file(s).** Save your final model files alongside the notebook.
- **Final notebook (`TeamName_FinalNotebook.ipynb`).** Retain the cells used for label construction, preprocessing, training, and evaluation. Add a final cell that loads the saved models, demonstrates inference for Task 1 and Task 2A, and clearly prints the inputs and predictions.
- **Peak-day allocation.** Include `submission_task2b.csv` and your written prioritization policy.
- **Prediction files submissions.** Include `submission_task1.csv` and `submission_task2a.csv`, using the exact columns and identifiers in the supplied templates.
- **Demo video.** Submit an unlisted YouTube video lasting three to five minutes. Explain your model architecture, preprocessing, label construction, and the challenges you encountered.
- **AI tool disclosure.** Explain which work was AI-assisted, which was not, and how you used the tools.

Judging criteria

Criterion	Weight
Data wrangling and label construction	20%
Model and architecture implementation	25%
Performance score (Task 1, task 2A)	20%
Task 2B allocation feasibility and prioritization policy	15%
Creativity of the solution	10%
Demo video	10%

Submission

Place all deliverables in one folder, compress it as `TeamName_Datathon.zip`, and upload it through the submission form.

Submit by **Friday, October 9, 2026, at 11:59 PM** Sri Lanka time (Day 15).

Deadline for submissions: **Friday, October 9, 2026, at 11:59 PM**

➤ **Submission Form:** <https://forms.gle/CcPPmttWdQgHvUdi6>

Datathon data reference

The supplied files represent records that an operational system would collect over time. Apply the following conventions across the datasets.

- Clock times use HH:MM in Asia/Colombo. Columns ending in `_time` contain clock times.
- Durations are measured in minutes. Columns ending in `_duration_min` contain durations.

Training files

File	Purpose
<code>deliveries_train.csv</code>	Each row is one order, identified by <code>delivery_id</code> . Each dispatched order is delivered as its own stop where its <code>route_id</code> and <code>seq_in_route</code> match exactly one route leg.
<code>route_legs_train.csv</code>	One row per route leg, from the depot or previous outlet to the next outlet. Includes planned and actual departure, travel, arrival, and completion times. Actual times appear only in these route records.

Test inputs

File	Purpose
<code>task1_test_inputs.csv</code>	Task 1 orders for a later period, with the order-record columns described below. All were dispatched.
<code>route_legs_test.csv</code>	Route legs associated with the Task 1 orders, containing planned rather than actual times.
<code>task2a_test_inputs.csv</code>	One row per depot, brand, and forecast week.
<code>task2b_peak_day_scenarios.csv</code>	One row per order in the peak-day scenario, with the information needed to allocate or defer it.
<code>task2b_peak_day_fleet.csv</code>	Vehicle availability for the peak-day scenario.

Reference tables

File	Purpose
<code>outlets.csv</code>	Outlet brand, district, depot, physical access, and delivery windows.
<code>vehicles.csv</code>	Vehicle type, temperature capability, capacity, fuel profile, and depot.
<code>calendar.csv</code>	Calendar context
<code>district_travel.csv</code>	District travel distances and free-flow times.
<code>service_allowance.csv</code>	The dispatcher's standard handling-time allowance for each brand and dock type.
<code>traffic_speed.csv</code>	Typical congestion by district and hour.
<code>road_conditions.csv</code>	Date-specific district disruptions, such as roadworks, flooding, or incidents.

Submission templates

Use the exact filenames below. Keep all supplied identifiers and rows unchanged. Task 1 also requires the original row order. Fill the answer columns and replace any placeholders. Identifiers match your submission to the correct records; mismatched identifiers cannot be scored.

File	Task
<code>submission_task1.csv</code>	Service time and lateness predictions
<code>submission_task2a.csv</code>	Weekly depot demand forecasts
<code>submission_task2b.csv</code>	Peak-day allocation

Order record columns

`deliveries_train.csv` and `task1_test_inputs.csv` · One row per order

Column	Format	Meaning
<code>delivery_id</code>	string	Unique order identifier and Task 1 prediction key.
<code>order_date</code>	date	The date the store's order was for.
<code>dispatch_date</code>	date	Date the order was dispatched. Blank if it never ran.
<code>dispatch_status</code>	string	<code>attempted</code> : dispatched on <code>order_date</code> . <code>deferred</code> : dispatched later because fleet capacity was short. <code>not_run</code> : never dispatched.
<code>outlet_id</code> , <code>brand</code> , <code>district</code> , <code>depot</code>	string	Outlet information
<code>temp_requirement</code>	string	chilled requires a refrigerated vehicle; ambient does not.
<code>order_units</code>	integer	Number of items or cases.
<code>order_weight_kg</code>	number	Order weight in kilograms.
<code>order_volume_m3</code>	number	Order volume in cubic meters.
<code>route_id</code>	string	Route carrying the order. Blank if it never ran.
<code>seq_in_route</code>	integer	Route position, starting at 0. Together with <code>route_id</code> this identifies the matching leg in the route records
<code>vehicle_id</code> , <code>vehicle_type</code> , <code>vehicle_temp</code>	string	Vehicle assigned to the order.
<code>planned_arrival_time</code>	HH:MM	Planned arrival time.
<code>window_open_time</code>	HH:MM	Start of the outlet's requested delivery window.
<code>window_close_time</code>	HH:MM	End of the outlet's requested delivery window.

Route leg columns

`route_legs_train.csv` and `route_legs_test.csv` · One row per route leg

Column	Format	Meaning
<code>leg_id</code>	string	Unique route-leg identifier.
<code>date</code>	date	Date the route ran.
<code>route_id</code>	string	Route identifier.
<code>seq</code>	integer	Position on the route, starting at 0.
<code>depot</code> , <code>vehicle_id</code> , <code>vehicle_type</code> , <code>vehicle_temp</code> , <code>brand</code> , <code>district</code>	string	Route attributes. A route serves one depot, brand, and district.
<code>from_point</code>	string	DEPOT for the first leg; otherwise the previous outlet.
<code>to_outlet</code>	string	Outlet served by this leg.
<code>distance_km</code>	number	Road distance in kilometers.
<code>planned_depart_time</code>	HH:MM	Planned departure from the previous point.
<code>planned_travel_duration_min</code>	minutes	Planned travel duration under clear-road assumptions.
<code>planned_arrival_time</code>	HH:MM	Planned arrival at the outlet.
<code>actual_depart_time</code>	HH:MM	Training only. Actual departure from the previous point.
<code>actual_travel_duration_min</code>	minutes	Training only. Actual travel duration.
<code>arrival_time</code>	HH:MM	Training only. Actual arrival at the outlet.
<code>leave_outlet_time</code>	HH:MM	Training only. Time the vehicle left after completing the delivery.
<code>monsoon</code>	0 or 1	1 for a monsoon or inter-monsoon month; otherwise 0.
<code>dow</code>	0 to 6	Day of the week, with 0 representing Monday.

Outlet columns

outlets.csv

Column	Format	Meaning
outlet_id	string	OUT001 through OUT120.
brand	string	Fresh, Style, or Tech.
district	string	One of 12 districts.
depot	string	Peliyagoda or Kandy, according to the outlet's assigned depot.
dock_type	string	rear_dock : a proper loading bay. street : curbside unloading. mall_bay : shared mall bay.
parking_constraint	string	normal : any vehicle. van_only : trucks cannot access the outlet. mall_dock : access is limited to the mall's delivery window.
mall_window	HH:MM-HH:MM	Fixed mall delivery window. Blank for outlets outside malls.
window_open_time	HH:MM	Start of the requested delivery window.
window_close_time	HH:MM	End of the requested delivery window.

Vehicle columns

vehicles.csv

Column	Format	Meaning
vehicle_id	string	VEH001 through VEH060.
type	string	truck or van. Only vans may serve van_only outlets.
temp	string	reefer : refrigerated and able to carry chilled or ambient goods. ambient : cannot carry chilled goods.
weight_cap_kg	number	Maximum weight per trip, in kilograms.
volume_cap_m3	number	Maximum volume per trip, in cubic meters.
fuel_type , km_per_l , weekly_fuel_quota_l	string, number, number	Fuel type, kilometers per liter, and weekly fuel quota in liters.
depot	string	Home depot. The vehicle serves only that depot's outlets.

Calendar columns

calendar.csv

Column	Format	Meaning
date	date	One row per calendar date.
dow, dow_name, is_weekend	0 to 6, string, 0 or 1	Day number, day name, and weekend flag. 0 represents Monday in dow.
iso_year, iso_week	integer	ISO year and week used to group demand.
is_payday	0 or 1	1 on payday.
festival	string	Festival name on the festival date; otherwise blank.
festival_ramp	0 to 1	Proximity to a festival, rising over the preceding nine days. 0 away from the festival; 1 on the festival date.
is_holiday	0 or 1	1 on a festival date or public holiday; otherwise 0.
monsoon	0 or 1	1 in a monsoon or inter-monsoon month; otherwise 0.
is_operating	0 or 1	1 when deliveries operate on that date; otherwise 0.

District travel columns

district_travel.csv

Column	Format	Meaning
district, depot	string	District and its serving depot.
road_class	string	urban, suburban, highway, or hill.
free_flow_kmh	km/h	Clear-road speed assumed in the plan.
depot_to_district_km	km	Distance from the depot to the district.
depot_to_district_freeflow_min	minutes	Clear-road travel time from the depot to the district.
inter_stop_km	km	Typical distance between outlets within the district.
inter_stop_freeflow_min	minutes	Typical clear-road travel time between outlets within the district.

Service allowance columns

`service_allowance.csv` · One row per brand and dock type

Column	Format	Meaning
<code>brand , dock_type</code>	string	One row per brand-and-dock-type combination.
<code>service_allowance_min</code>	minutes	Handling time budgeted per stop. This is a planning allowance, not an observed duration.

Traffic speed columns

`traffic_speed.csv` · Documented key columns

Column	Format	Meaning
<code>monsoon</code>	0 or 1	1 in a monsoon or inter-monsoon month; otherwise 0.
<code>speed_index</code>	number	Speed relative to clear roads. 100 means free-flowing traffic; lower values mean slower travel.

Road condition columns

`road_conditions.csv` · Documented key columns

Column	Format	Meaning
<code>disruption_index</code>	number	Date-specific disruption in a district. 100 means clear conditions; lower values indicate disruption such as roadworks, flooding, or an incident.

Peak day order columns

task2b_peak_day_scenarios.csv

Column	Format	Meaning
scenario	string	Always S1.
order_ref	string	Unique order identifier within the scenario. Use this as the allocation key.
outlet_id , brand , district , depot	string	Outlet and network identifiers for the order.
dock_type , parking_constraint , mall_window , window_open_time , window_close_time	string	Outlet access and delivery-window information copied into the scenario file.
temp_requirement	string	chilled or ambient.
order_units , order_weight_kg , order_volume_m3	number	Order size in units, kilograms, and cubic meters.
deferred_yesterday	0 or 1	1 if the outlet was skipped on the previous run.
days_since_last_served	integer	Days since the outlet last received a delivery.

Peak day fleet columns

task2b_peak_day_fleet.csv

Column	Format	Meaning
scenario	string	Always S1.
vehicle_id , status	string	Vehicle identifier and availability. Available vehicles can be allocated; in_workshop vehicles cannot.

Validation scripts

check_allocation.py checks the Task 2B allocation against the feasibility rules and can be run before you submit. Passing these checks confirms that your allocation meets the feasibility rules, not that it is optimal. Judges assess how you prioritized orders and explain your deferrals.

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Submission links

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GOOD LUCK!